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Proceedings, Part I

Part 1

Nguyen Thai-Nghe · Thanh-Nghi Do ·
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Automatic Segmentation of Masses on Mammograms Using Fuzzy Logic

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Abstract. Breast cancer is the leading cause of death among women globally. Early detection of abnormalities is crucial for enhancing treatment effectiveness. Mammography, the most common screening tool for breast cancer, presents challenges due to the manual interpretation of mammograms and variability among radiologists. In this paper, we propose an automatic breast mass detection system. This system utilizes fuzzy logic for image preprocessing, in conjunction with deep learning network models like U-Net and Mask R-CNN for mass segmentation. Our experimental results indicate that this method improves the quality of input images, efficiently retains crucial features, and aids in accurately localizing masses. It achieved a high accuracy rate of 92.5% on the data set collected from several hospitals. These results serve as a significant foundation for our ongoing development of fuzzy learning techniques in classifying and identifying masses in mammograms.

Keywords: Breast cancer · mammogram · fuzzy logic

1 Overview

Breast cancer (BC) predicted has affected 2.3 million women worldwide in 2020, with 685,000 fatalities. Breast cancer had been detected in 7.8 million females in the previous five years as of 2020, making it the most common cancer on the planet. It causes more disability-adjusted life years (DALYs) in women worldwide than any other form of cancer [1].

Lévy et al. [2] focuses on the development and evaluation of a deep learning-based system for classifying breast masses in mammograms. The primary goal is to enhance the diagnostic accuracy of mammograms by using CNNs to classify breast masses as benign or malignant. The authors utilized a deep CNN architecture trained on a large dataset of annotated mammogram images. The network was designed to learn intricate patterns and features associated with different types of breast masses through multiple layers of convolutions, activations, and pooling operations. The study leverages the capabilities of deep

convolutional neural networks (CNNs) to automatically identify and categorize suspicious areas within mammographic images, aiming to improve the accuracy and efficiency of breast cancer detection. The CNN model developed by Lévy achieved a high classification accuracy up to 92.9% on DDSM dataset. Agarwal et al. [3] presents a detailed study on the application of deep learning techniques for the automated detection of breast masses in mammographic images. The primary focus of the research is to develop a system that leverages the capabilities of deep convolutional neural networks (CNNs) to enhance the accuracy and efficiency of breast cancer detection. To improve the model's performance, extensive preprocessing steps are applied to the mammogram images. These steps include normalization, noise reduction, and data augmentation to ensure the model can generalize well to new, unseen data. The proposed CNN architecture consists of multiple convolutional layers interspersed with activation functions (such as ReLU) and pooling layers to progressively distill the important features from the input images. The final layers of the network are fully connected, leading to a classification output that indicates the presence or absence of a mass. The high accuracy 91% of the CNN model highlights its potential to improve the early detection rates of breast cancer, which is crucial for successful treatment and patient survival. The article by Shen et al. [4] explores the application of deep learning techniques to enhance the accuracy and efficiency of breast cancer detection in mammographic screenings. The research highlights the potential of deep convolutional neural networks (CNNs) to revolutionize the screening process by providing more reliable and consistent interpretations of mammograms. The study includes rigorous preprocessing steps to prepare the mammogram images for training the CNN. This includes normalization to standardize the images, noise reduction techniques to enhance image quality, and data augmentation to increase the diversity of the training set and prevent overfitting. The deep learning model developed by Shen demonstrated significant improvements in the detection accuracy of breast cancer compared to traditional methods and manual readings by radiologists. The test set of full-field digital mammography (FFDM) images from the INbreast database achieved a per-image AUC of 0.95. Dezso Ribli's research [5] demonstrates the significant advantages of utilizing deep learning techniques for detecting and classifying breast mass in mammograms. The study emphasizes the potential of deep convolutional neural networks (CNNs) to accurately identify and categorize mass, which is crucial for the early diagnosis and treatment of breast cancer. The author propose a CAD system based on one of the most successful object detection frameworks, Faster R-CNN with accuray archived 95% on the public INbreast database (Fig. 1).

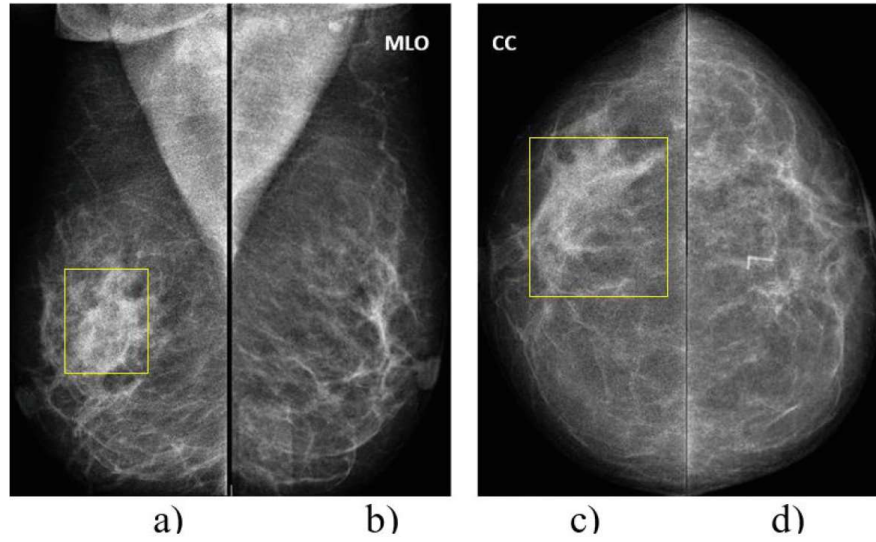


Fig. 1. Images on 2 views a) RMLO b) LMLO c) RCC d) LCC with the mass location localized in pictures a) and c)

2 Background

2.1 Neural Network Models

The architecture of Convolutional Neural Networks (CNN) is commonly utilized in the field of computer vision. It's viewed as a solution to optimize network training processes by reducing weights, which in turn increases computational speed and decreases training time. The basic CNN architecture includes the convolution layer, nonlinear layer, pooling layer, and fully connected layer, all connected in a specific order. Typically, input data are first processed through the convolutional and nonlinear layers, then the resulting values are passed through the pooling layer. The sequence of the convolution layer, nonlinear layer, and pooling layer is repeated multiple times within the network. Finally, the result is sent through the fully connected layer for classification (Fig. 2).

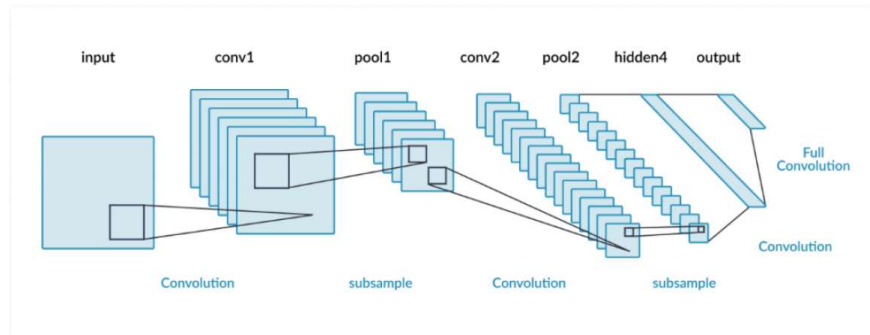


Fig. 2. An illustration of CNN network architecture [9]

2.2 U-Net

U-Net, developed by Ronneberger et al. [8], is primarily used for image segmentation. Its architecture is depicted in Fig. 3. U-Net is a symmetric network comprising two paths: contraction and expansion. The contraction path, or the encoder on the left block, resembles a standard convolution network and provides classification information. The expansion path, or the decoder on the right block, includes deconvolution layers for learning and segmentation. To prevent information loss during convolution and deconvolution, short connections link two blocks at the same level. The contraction path extracts features from the image for learning, and reduces the image size using convolutions and pooling layers to generate high-resolution feature maps. The expansion path allows the network to learn localized classification information.

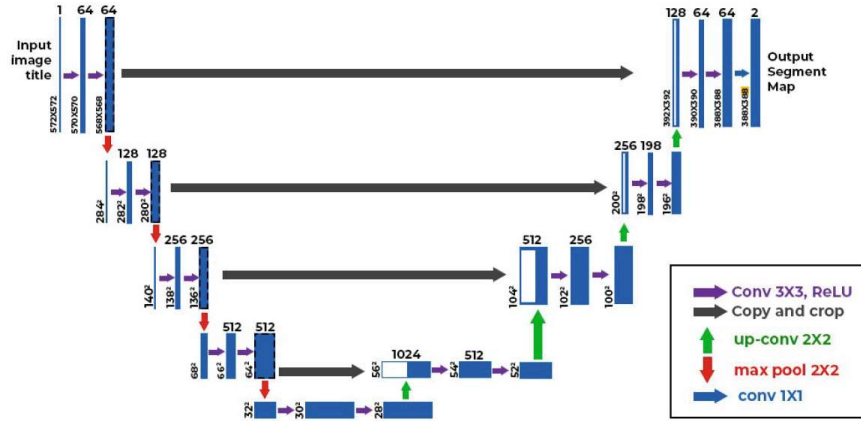


Fig. 3. U-net network architecture [8]

2.3 Mask R-CNN

Mask R-CNN [11] is a high-performance object classification model based on pixel-level image segmentation. Mask R-CNN adds a branch to predict the object's mask in parallel with the current branches. The Loss function of Mask R-CNN is a combination of 3 Loss: classification, localization and segmentation mask.

2.4 Fuzzy Logic

Fuzzy logic [6] is a method of processing information and making decisions based on ambiguous or uncertain values. Unlike classical logic, which uses only binary values (0 or 1), fuzzy logic allows for intermediate values between 0 and 1, thereby describing states that are not strictly true or false. Fuzzy machine learning integrates these fuzzy learning techniques into machine learning [7]. These models

are particularly effective for handling uncertain information often found in medical images, facilitating inference and analysis of abnormal image regions, and thereby increasing efficiency in diagnosis and classification. A fuzzy inference system (see Fig. 4) allows decision-making based on fuzzy rules and a knowledge base.

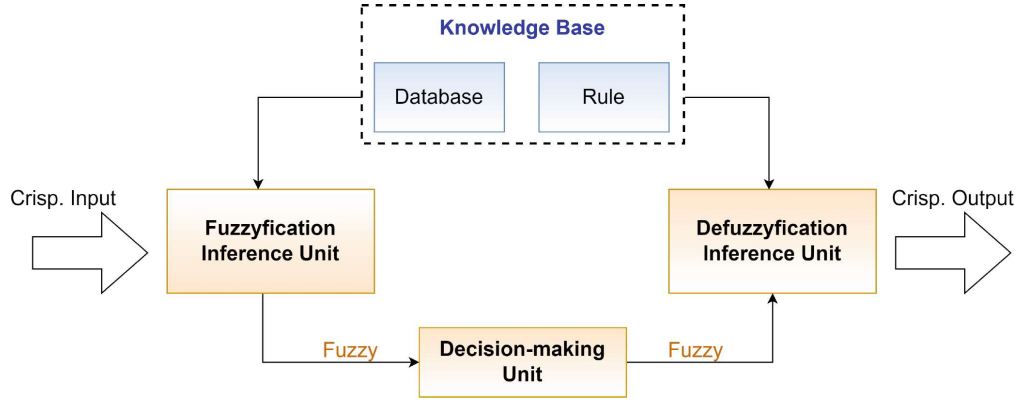


Fig. 4. Fuzzy inference system (FIS) [10]

2.5 Evaluation Metrics

In segmentation models like U-Net and Mask R-CNN, the loss measurement formula is typically designed to reflect the discrepancy between the predicted outcome and the actual label. We use loss and accuracy measurements based on the following formula:

$$Accuracy = \frac{TP + TN}{TP + TN + FP + FN} \quad (1)$$

$$Dice\ Coefficient = \frac{2 \times TP}{2 \times TP + FP + FN} \quad (2)$$

$$Dice\ Loss = 1 - Dice\ Coefficient \quad (3)$$

In which: True Positive - TP: true positive, False Positive - FP: false positive, True Negative - TN: true negative, False Negative - FN: false negative.

3 Proposed Method

In this paper, we propose a three-stage general model. The stages include: (1) preprocessing to create input data for a deep learning model using fuzzy logic, (2) feature extraction and training, and (3) mass detection and localization. The proposed model is presented in Fig. 5.

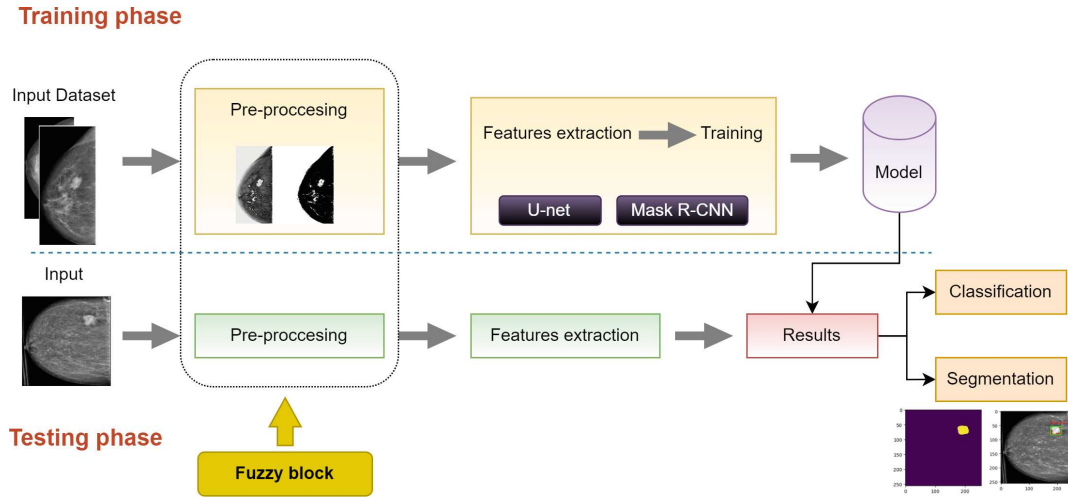


Fig. 5. The proposed general model detects and localizes mass on mammograms.

3.1 Pre-processing Data Using Fuzzy Logic

Input images will be standardized, stored in .jpg format, and set to dimensions of 256×256 . With a doctor's assistance, we'll identify damaged areas based on variations in density on mammograms and classify them. Image preprocessing is performed on the input data by normalizing the image size and enhancing the image using histogram, Gaussian, and Gompertz algorithms (Fig. 6). The output image then serves as the input for the fuzzy logic system. By running the image through the fuzzy inference system shown in Fig. 7, we retain attenuation regions that include the mass area on the mammogram. Information such as location and coordinates is then saved for input into the feature extraction process using machine learning models.

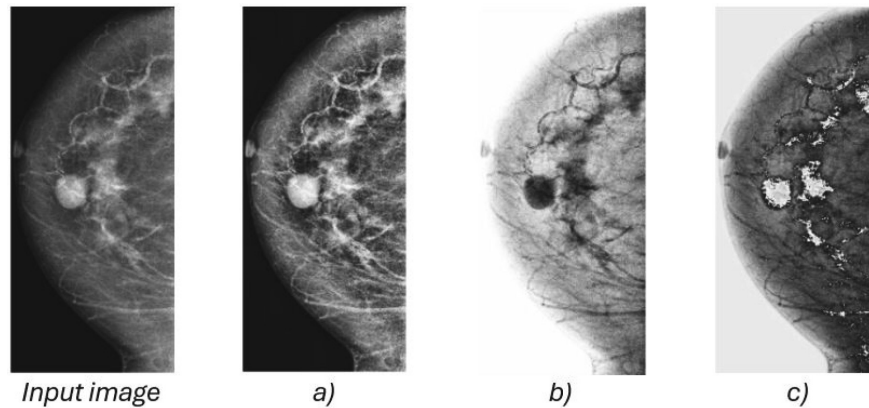


Fig. 6. Output results after filtering the image using the filters a) Histogram b) Gaussian c) Gompertz

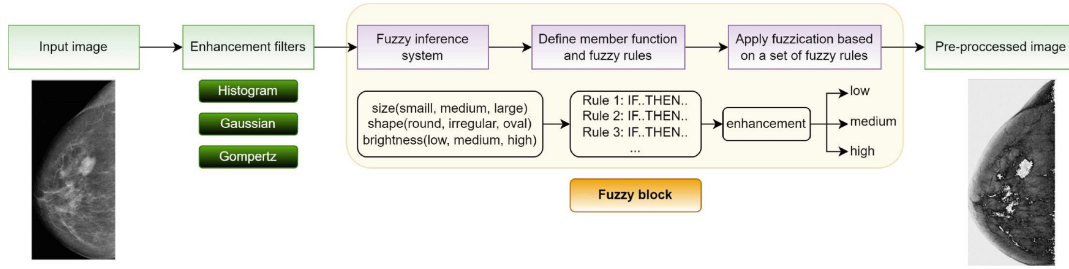


Fig. 7. The preprocessing of the input image applies fuzzy logic

Steps to perform image preprocessing using fuzzy logic techniques:

Step 1. Preprocessing and enhanced filtering: We use three methods including.

- + Histogram method with a threshold of 200.
- + Gaussian function with values $a=0.8$ and $b = 4$.
- + Gompertz function with values $c=3$ and $b = -0.5$.

Step 2. Fuzzy block:

+ Fuzzy Inference System: This system determines the level of mass enhancement in medical images based on three main factors: size, shape, and brightness.

+ Fuzzy Member Functions: These describe the levels of each fuzzy variable. Values include size (small, medium, large), shape (round, irregular, oval), and brightness (low, medium, high).

+ Apply Fuzzy Rules: This creates output images with mass-containing areas retained. Pixels within these areas are kept. The set of fuzzy rules applied in this document is as follows:

- Rule 1: If the size is 'large' and the shape is 'round' or 'oval' and the brightness is 'high', then the enhancement level is 'high'.
- Rule 2: If size is 'medium' and shape is 'round' or 'oval' and brightness is 'medium', then the enhancement level is 'medium'.
- Rule 3: If size is 'small' and brightness is 'low', then the enhancement level is 'low'.

Step 3. Perform segmentation on bright areas, eliminating small bright areas based on a given threshold.

Step 4. Use mapping to convert binary images to grayscale images.

3.2 Feature Extraction

We improved a method by constructing a model to detect and classify masses on mammograms. This model was implemented on U-net and Mask R-CNN networks. The process starts with preprocessing the input image through an inference system and fuzzy analysis. After preprocessing, the images undergo feature extraction, detection, and classification. During feature extraction, we use weights from previously trained models, applying transfer learning techniques.

We then continue to fine-tune the weights to decrease the Loss values. Training stops when the Loss value no longer improves. Ultimately, four trained models corresponding to the input data set are obtained.

3.3 Detection and Classification of Masses

Following feature extraction, the dataset images are processed by four trained models to produce corresponding results. Key output values include images, boxes, classes, and scores, with scores being particularly important as they're used to determine the final classification result. The compiled results are then filtered through the classification layers, prioritizing the highest value scores. Upon completion of the classification process, the output image will display both a bounding box and the probability of recognition.

4 Experiment Results

4.1 Applicable Scenarios

For the proposed model, we conduct experiments under four scenarios with the following training parameters (Table 1):

Table 1. Recommended scenarios and training parameters.

Scenario	Name	Preprocessing data	Learning rate	Epoch	mAP@IoU	Num classes
1	Mask R-CNN	—	0.0001	100	0.5	2
2	U-Net	—	0.0001	100	0.5	2
3	F-Mask R-CNN	Fuzzy block	0.0001	100	0.5	2
4	F-U-Net	Fuzzy block	0.0001	100	0.5	2

In scenarios 1 and 2, we utilize Mask R-CNN and U-Net networks for feature extraction and training. For scenarios 3 and 4, we apply fuzzy logic in the image preprocessing stage, as presented in Sect. 3. After conducting experiments with various sets of parameters, we selected identical parameters for all four scenarios. The learning rate is set at 0.0001, with the training being conducted 100 times. We used the $\text{mAP@IoU} = 0.5$ parameter, with the number of classes (Num classes) being 2, corresponding to two types: mass and non-mass.

4.2 Installation Environment and Dataset

- Installation Environment: The system runs in a Python environment on Kaggle, using an Intel®Xeon®2.00 GHz processor with 30 GB RAM. It also utilizes a graphics processor with a 16 GB RAM configuration and the Nvidia Tesla P100 GPU. Tensorflow v1.5 is the library used for network model training.

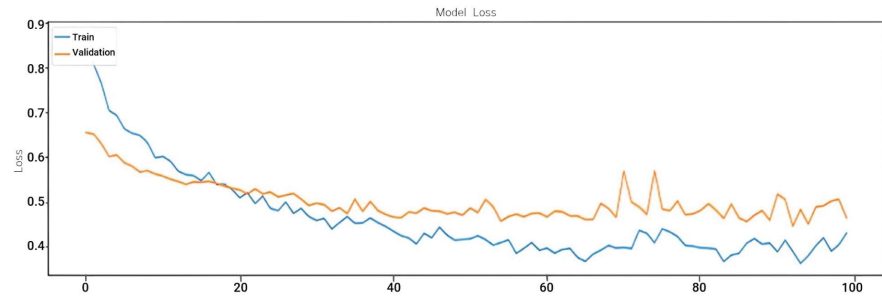
- Experimental Dataset: The data set was compiled from the VinDr dataset, supplemented by data from several hospitals, and subsequently reviewed by doctors who localized the masses. After preprocessing the slices containing the mass and converting them to .jpg format, the dataset for training and testing comprises 7990 images. Of these, 6392 images constitute the training set, and 1598 images form the testing set, maintaining an 80/20 split. After classifying by mass, the dataset includes 7327 normal images without masses, while 663 images have been labeled and localized by doctors.

4.3 Training Results

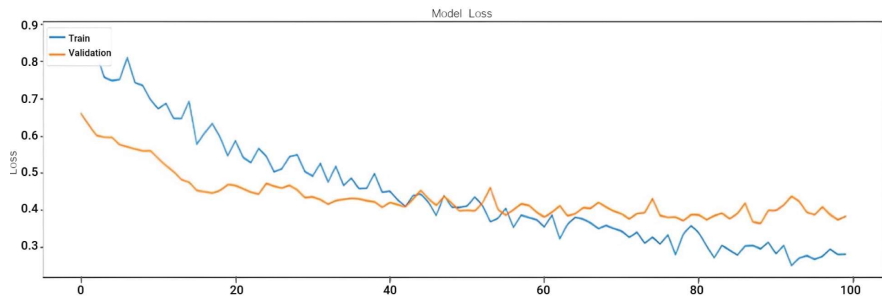
In this section, we present the results from our experiments. These include the measurement of Loss and Accuracy values for the proposed scenarios, training and testing times, as well as mass detection and classification outcomes for the proposed scenarios.

Refer to Fig. 8 for a chart displaying the model training over four scenarios. These training sets include the mask of the mass. The model effectively extracts features and learns the mass’s characteristics, enhancing the learning capacity of each model through fuzzy logic techniques applied in image preprocessing. The models extract general features based on the mass’s characteristics in the image. Training will stop when the Loss value no longer decreases, at which point we save the best model. The average TotalLoss values measured at this point are 0.4015, 0.3150, 0.1503, 0.1013 respectively. Figure 9 shows the accuracy of 4 scenarios. The results show that scenarios 3 and 4 after applying fuzzy logic in image preprocessing give better accuracy results than scenarios 1 and 2.

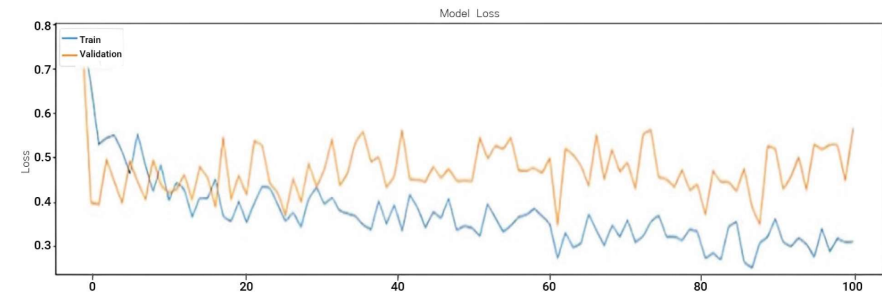
The training spans over 100 epochs to observe the model’s stability. To prevent data overfitting, we implement additional techniques such as dropout and early stopping, which help the model maintain stability and efficiency when conducting tests. The total training time and accuracy of the four scenarios are presented in Fig. 10. In scenarios 1 and 3, the training time using Mask R-CNN is considerably longer compared to the U-net model.



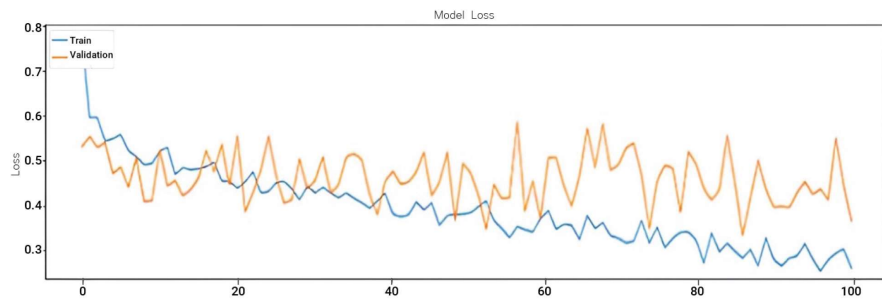
(a) Scenario 1



(b) Scenario 2



(c) Scenario 3



(d) Scenario 4

Fig. 8. Loss of 4 scenarios respectively

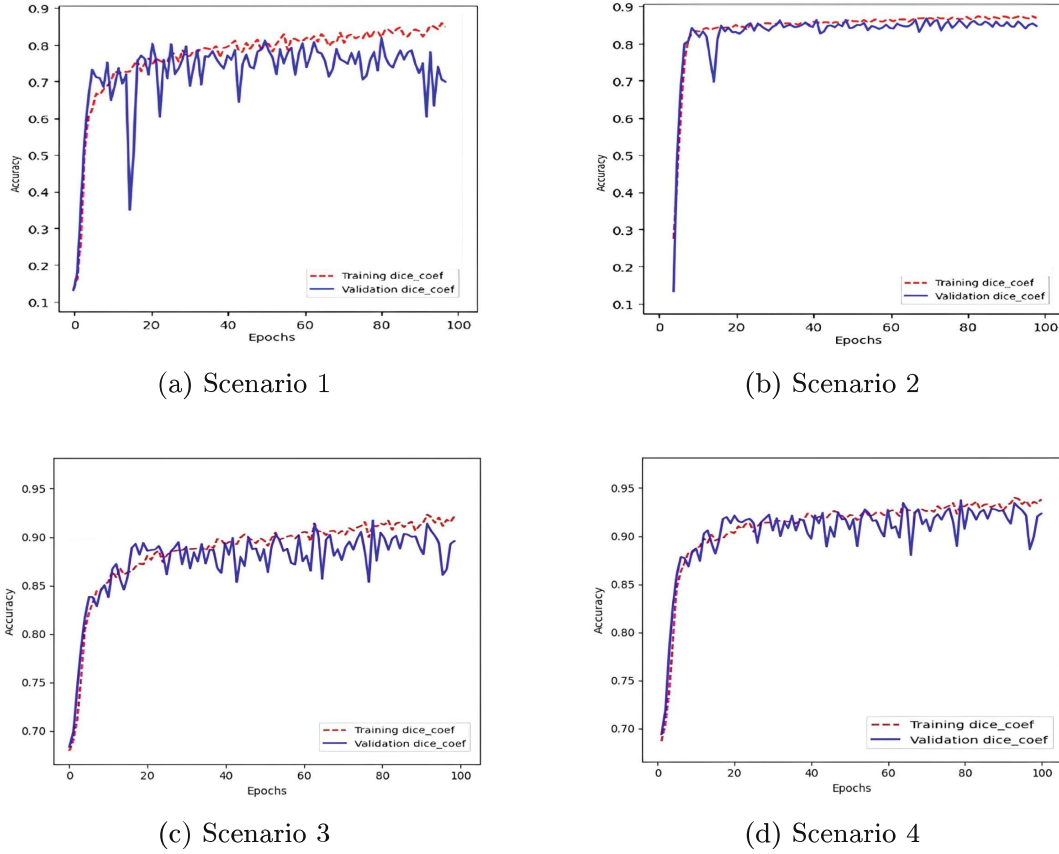


Fig. 9. Accuracy measurement in 4 scenarios respectively

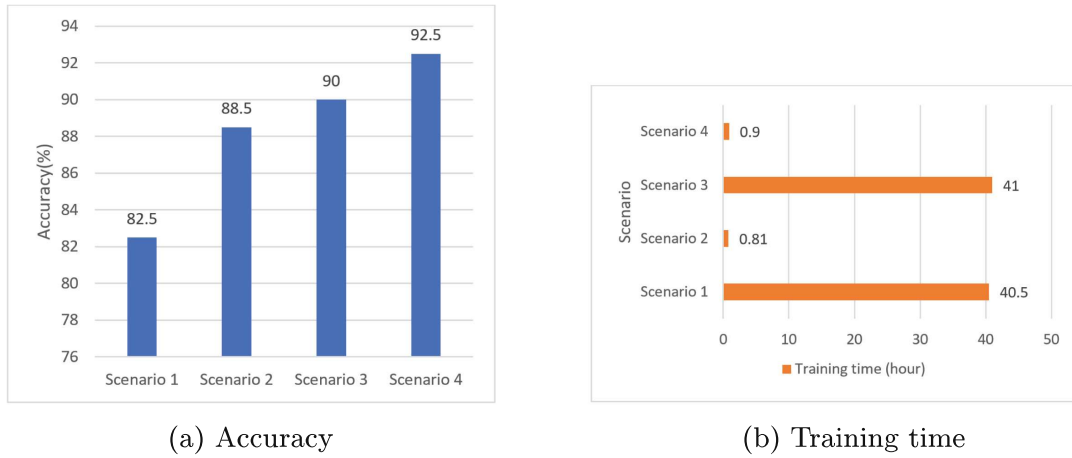


Fig. 10. Accuracy and training time for 4 scenarios.

4.4 Results of Mass Detection and Localization

To objectively evaluate the results, we use a separate dataset from the training set to test the classification and segmentation capabilities. The original masked images, labeled by the physician and marked with mass segmentation, were pro-

cessed through four prediction models. The results, including the segmentation outcomes from the proposed models and the localization results with corresponding accuracy, are displayed in Fig. 11.

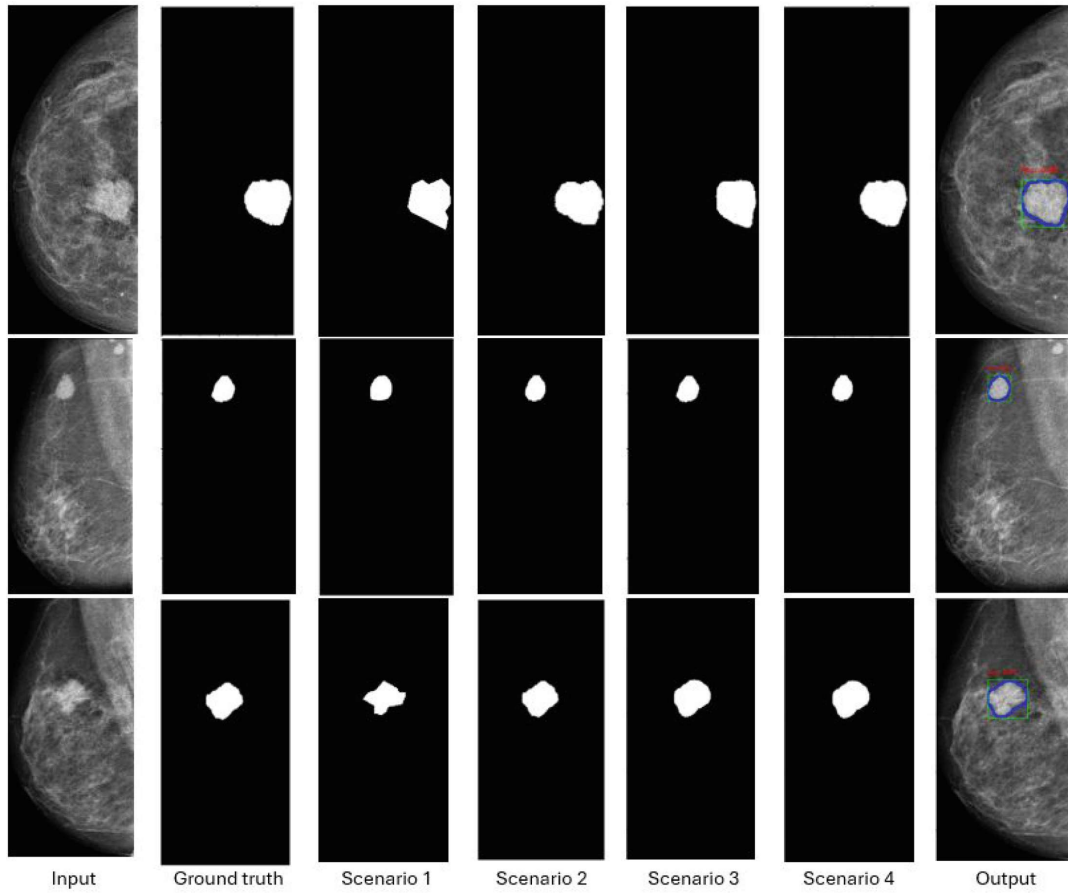


Fig. 11. Mass prediction results on 4 scenarios. Result image with localized mass on scenario 4.

As shown in Fig. 11, the location of the mass area in the mammogram is determined using a prediction mask that aligns with the original image mask. Even images with masses of complex shapes demonstrate high performance, with an accurate identification rate exceeding 90%. This represents a positive outcome, as it effectively aids doctors during the diagnostic and treatment process.

4.5 Compare and Evaluate the Proposed Model

Through the process of training the proposed models, the comparison criteria are summarized as Table 2:

Table 2. Model evaluation measurement results

Scenario	Loss	Training time (hours)	Accuracy (%)
Scenario 1	0.4015	40.5	82.5
Scenario 2	0.3150	0.81	88.5
Scenario 3	0.1503	41	90
Scenario 4	0.1013	0.9	92.5

The outcomes from the four scenarios illustrate the effectiveness of using fuzzy logic techniques on input images to enhance identification accuracy. Even though the improvement rate isn't substantial, the research findings indicate a promising method for early detection and diagnosis of abnormalities in mammograms.

5 Conclusion

In this paper, we proposed four models for detecting and segmenting masses in mammograms. These models achieved an accuracy of 90% with the Mask R-CNN network and 92.5% with the U-Net network. We also introduced a method to increase the classification model's accuracy. This method uses fuzzy logic techniques to enhance the initial image quality before feeding it into the classification model, thus improving the classification results. However, this study has certain limitations. The detection and classification were only performed on masses, without considering microcalcification areas in mammograms. In the future, we intend to explore additional methods to boost model accuracy and decrease model training time. Incorporating more significant features will aid in detecting and classifying benign and malignant masses.

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